Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Sovereign Collaborative Energy Usage Optimization			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Knowledge Services	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Power IT	Intelligent Power Grid Platform Technologies	R&D / Engineering Services	Intellectual Property Analysis / Management Technologies
Concept of Proposed Technology	The global energy transition is driving a fundamental shift from centralized power generation to distributed renewable energy sources. Combined with a drastic surge in electricity demand driven by AI research, digital transformation, and increased industrial activity, this transition has resulted in severe network congestion.			

In Korea specifically, where 84% of energy is imported[1], the urgency for renewable energy adoption is compounded by an increased demand for strategic autonomy[2]. However, to achieve these future goals, policy makers must ensure sufficient support for this transition. Given that the Korean economy relies heavily on energy-intensive exports[3], this support risks eroding if no action is taken to help energy-demanding organizations maintain consumption levels despite the paradigm shift.

A core problem driving network congestion is that within industrial parks, commercial clusters, and office complexes, multiple energy-intensive activities often occur simultaneously – for instance, during daily startup operations. This not only creates unnecessary network stress but also fails to optimally utilize available sustainable energy sources. In itself this is wasteful, but it also results in unused renewable energy that may overflow the network later in the day, causing additional stress.

Our solution addresses these challenges through collaborative optimization that brings organizations together to spread demand across the fluctuating daily supply curve, stay within the electricity network's capacity limits, and ensure fossil fuels are only used when sustainable sources cannot meet demand.

Achieving this optimization requires organizations to execute a collaborative algorithm that identifies when the collective group fails to effectively utilize available renewable energy resources and when they collectively overdemand grid capacity. This collaborative algorithm necessitates organizations sharing energy consumption data. This data is typically highly sensitive as it acts as an operational fingerprint, exposing production volumes, load profiles and product scale-ups.

By using data spaces, we ensure this data will be handled sovereignly and securely in line with the agreements that will be made amongst participants. On top of that, we will deploy the optimization algorithm in the form of a multi-party computation algorithm. Through cryptographic means, this technique ensures that no party other than the data owner can inspect the raw data while still enabling a valid and fair energy usage division. The combination of data spaces with multi-party computation will ensure that no (accidental) spill of intellectual property can occur. Furthermore, data spaces are designed to be provider-agnostic. This means that users of the algorithm will be able to select from multiple IT service providers. As such, we are setting up the solution with scalability in mind.

That said, even when confidentiality is technically guaranteed, organizations may be hesitant to participate in data sharing solutions because they do not trust the solution because of it being

[1] https://www.iea.org/countries/korea/energy-mix

[2] https://cms.apln.network/wp-content/uploads/2024/05/Joon-Hyung-Kim_27-May.pdf

[3] https://www.iea.org/countries/korea/energy-mix

	overcomplex. To address this barrier, we integrate AI assistance during the establishment of the optimization agreement. This will help the user to create a thorough understanding of what will be shared and how it will be shared. Furthermore, the assistance can alleviate repetitive negotiations by extracting user preferences as contracts are being negotiated. These preferences can be tested against the user. This results in an increase in support and adoption rate for energy optimization solutions whilst reducing the total set-up time.
Rationale for Joint R&D Project	Europe and Korea face remarkably similar challenges in energy network management: • Energy network congestion due to the transition to distributed and unpredictable renewable energy sources • Dramatically increased energy demand driven by AI research and digital transformation • Rising electricity costs creating bottlenecks for industrial scaling • Energy security concerns amplified by geopolitical instability • Electricity price spikes and supply uncertainty directly impacts the competitiveness and scaling potential of Korean and European SMEs Korea has pledged to become carbon neutral by 2050, which requires innovative approaches to energy efficiency. At the same time, a growing energy demand, high energy import rate, and the slow pace with which qualitative improvements are made on the national grid, all cause energy prices to increase significantly. This impacts SMEs specifically in a disproportionate fashion; at the same time, Korea's modern industrial parks and smart city initiatives provide ideal environments to demonstrate innovative ways to collaboratively optimize energy use and provide a higher SME scaling potential. TNO is uniquely positioned to undertake a project in the energy domain using data space technology. TNO is global leader in the domain of data spaces. We have developed an open-source implementation of the International Data Spaces Association's reference architecture model (IDS-RAM), and are board members for, amongst others, the International Data Spaces Association (IDSA), Gaia-X, the Big Data Value Association (BDVA), and the AI, Data & Robotics Association (ADRA). We are the technical coordinator of the Data Spaces Blueprint of the Data Spaces Support Centre (DSSC), which we have successfully implemented in various sectors – including the energy sector. Through our involvement in the Hedge-IoT and Enershare, and nationally in Het Normo, we have built up knowledge on grid optimization and IoT-enabled energy ecosystems. Our expertise extends beyond infrastructure. Through our research on engineering principles for building trusted systems where humans and AI interact, we have demonstrated a deep understanding of human–AI ecosystems, particularly in how to design AI-based decision support systems. Finally, we are the lead of the Dutch Centre of Excellence for Data Sharing & Cloud (CoE-DSC), which positions us at the intersection of innovation and implementation and supporting national use-cases with their data sharing implementations.

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	This collaboration aligns with both Korean and European priorities in energy transition, digital sovereignty, and AI innovation. It demonstrates how privacy-preserving technologies can enable collaboration without sacrificing competitive confidentiality, addressing a fundamental challenge in the data economy.
Key Words	1. Network Congestion Mitigation 2. Data Spaces 3. Multi-party Computation 4. Data Sovereignty 5. AI Assisted Negotiation 6. Digital Contracts 7. Digital Trust

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (TRL 4)	**End of the Project**	Level (TRL 7)
R&D Objectives	The overall project objective is twofold: first, to optimize energy resource utilization by enabling collaborative consumption scheduling that maximizes use of existing network infrastructure and available sustainable energy sources, thereby accelerating the energy transition; second, to build digital trust through privacy-preserving multi-party computation and AI-assisted contract negotiation, reducing barriers to collaborative data sharing. A critical concern for organizations is protecting intellectual property. Through data sovereignty principles, organizations maintain complete control over their data, which never leaves their systems in raw form, ensuring no accidental IP disclosure through platform participation. Korean SMEs can use this technology to reduce electricity costs, remove energy network capacity as growth constraints, and participate in the broader data economy with confidence that their competitive information and intellectual property remain protected in an understandable way. From the technology perspective, this requires integration of three core capabilities into a unified platform: 1. data space infrastructure enabling sovereign provider-agnostic data exchange where organizations retain full control over their data; 2. privacy-preserving multi-party computation protocols computing optimal energy distribution without any party accessing others' raw operational data, preventing IP leakage; 3. AI-assisted negotiation systems automating complex multi-party contract establishment while clearly explaining IP protection guarantees so that the organization truly grasps what they are committing themselves to.			

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	Aside from AI-assisted negotiation systems, these constituent technologies have been demonstrated separately before. In this project, we will create and or adapt these technologies so that they are integrated, adapted to Korean energy legislation, market structures, and organizational contexts, and validate the integrated platform with several organizations in a Korean industrial cluster to reach TRL 7. The Korean SME partner will receive comprehensive knowledge transfer enabling independent deployment and commercialization across additional clusters and other data sharing domains beyond energy optimization.
R&D Target Specifications	The performance of the collaborative energy optimization platform will be evaluated using technical, user adoption, and operational metrics to demonstrate readiness for commercial deployment at TRL 7. Energy optimization performance will be measured through quantitative infrastructure metrics: peak load reduction assessed by comparing maximum simultaneous demand during baseline versus optimized scheduling periods; cost savings calculated by comparing aggregate electricity expenses across participating organizations before and after implementation; network utilization efficiency evaluated by analyzing smoothness of demand curves and alignment of consumption with sustainable energy supply. Data sovereignty and confidentiality preservation will be validated through systematic verification that the data that is transferred between organizations is only transferred in encrypted form, with complete audit trails documenting all data access requests, computation executions, and the division between AI-automated and human decisions in contract negotiations. User trust and adoption can be measured using validated survey instruments administered at project milestones. We can individually quantify user approval for trust in privacy-preserving mechanisms and IP protection; approval for contract clarity and understandability of data sharing agreements; reduction in contract negotiation time compared to traditional manual negotiation (baseline established through user surveys); and tracking of progressive automation adoption while maintaining user satisfaction with retained control. The platform will be demonstrated with several participating organizations within a Korean industrial cluster at TRL 7, with metrics collected through system telemetry (network load patterns, data access logs, computation traces), user surveys, and qualitative interviews, enabling quantitative comparison of baseline versus demonstration performance and proving commercial readiness for broader deployment.
R&D Plan	Phase 1: Foundation & Requirements (6 months) Aim: Establish partnership agreements, distill requirements from end users and Korean context, and deliver validated architecture design (TRL 4→5). Milestones

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	• Partnership agreement & scenario (M1, ≈Month 1): formalized TNO–Korean partner collaboration; identified target industrial cluster with committed participants; agreed demonstration scope and success criteria. • Requirements & architecture baseline (M2, ≈Month 3): documented technical and regulatory requirements based on Korean energy legislation, end user consumption patterns, and digital trust requirements; solution architecture designed (data space, MPC protocols, AI negotiation). • Data readiness & baseline measurements (M3–4, ≈Month 5): end user systems prepared for MPC integration; baseline electricity usage patterns established for validation. • Validated architecture & Phase 2 plan (M5, ≈Month 6): architecture validated with partner and end users; confirmed alignment on technical approach; finalized Phase 2 implementation plan. Phase 2: Development, Integration & Demonstration (30 months) Aim: Develop collaborative optimization and AI negotiation components, integrate with end user systems, and validate end-to-end solution in operational settings (TRL 7). Milestones • Experimentation data space (M6, ≈Month 8): functional mocked data space simulating end user connectors and data patterns; testing infrastructure for rapid iteration. • Collaborative energy optimization prototype (M7, ≈Month 17): secure MPC protocols implemented; fair energy distribution algorithms validated in mocked environment. • AI-assisted negotiation system (M8, ≈Month 26): NLP for contract explanation; preference extraction and automated negotiation mechanisms; progressive automation framework with human oversight validated. • Operational demonstration (M9, ≈Month 29): solution integrated with 5–10 organizations' actual systems; deployed with real energy data; continuous monitoring dashboards tracking optimization gains, peak load reductions, and cost savings; validated audit trails and data sovereignty; user feedback collected. • TRL 7 demonstrators & knowledge transfer (M10, ≈Month 30): evaluation report comparing baseline vs. demonstration results; technology transfer to Korean partner (documentation, training); academic publications; commercialization roadmap.

4. R&R (Roles and Responsibilities)

Expected R&R of TNO	• Lead applied research on data space architecture, privacy-preserving data sharing and multi-party computation, ensuring the process is undertaken with scientific rigor. • Develop and validate an AI-based method for automated contract negotiations for data sharing contracts. • Develop and validate a privacy-preserving distributed algorithm for energy use optimization.

	• Integrate the beforementioned data space technology, contract negotiation system, and MPC algorithmics into a single platform for continuous energy use optimization and monitoring, which can be readily connected external systems for in-the-field demonstration. • Design evaluation protocols and metrics for technical and functional performance of the developed components and platform, including the creation of a testbed environment with mocked data integrations. • Coordinate the scenario definition process ensuring its relevance to SME partners' use cases. • Support the adoption by SMEs and ecosystem partners by disseminating knowledge and best practices, e.g. through publications and workshops.
Expected R&R of Korean SME Partner	Phase 1 (6 months): Foundation & Requirements • Scenario definition: Define the concrete use case and industrial cluster to which the use case is applicable. • Data readiness: ensure that end user systems meet data quality standards for MPC integration. Align data formats and provide integration points for the MPC algorithm, optionally provide testing datasets for initial development and validation. • Regulatory & governance input: Advise on Korean energy legislation and contract law, market and organizational structures, along with requirements for digital trust. • Energy operations input: Advise on end user energy consumption patterns and applicable operational constraints. Phase 2 (24 months): Integration & Demonstration • Demonstration & validation: Integrate the developed platform with systems of several organizations; help shape and execute methodology for validating the performance of the solution; assist in the auditing for sustained data sovereignty and digital trust. • Compliance & trust: Ensure our innovations stay aligned with Korean energy legislation and contract law; provide feedback about the level of trust that the system evokes. • Relevance guardianship: Provide continuous feedback about the feasibility and maintained relevance of our innovation. • Dissemination & valorization: In collaboration with TNO, contribute to dissemination activities to promote (future) adoption; advise on valorization potential and potential strategies

5. Expected Outcomes/Impact

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Expected Outcomes and Impact	The outcomes of the project are: • A TRL 7 collaborative energy optimization platform integrating data space infrastructure, privacy-preserving multi-party computation protocols, and AI-assisted negotiation capabilities, demonstrated with several organizations in a Korean industrial cluster • Deployed data space infrastructure across participating end users, enabling scalable provider-agnostic participation and positioning organizations for broader data economy engagement • An AI-assisted digital contract negotiation system that explains complex multi-party agreements in accessible language, extracts organizational preferences, and automates negotiation while maintaining human oversight • Validated methodology and documentation enabling replication across additional industrial clusters and commercial complexes The impact of the project on the short term covers mainly effects on the organizational level, which are: • Reduced electricity costs through collaborative optimization smoothing demand curves and aligning consumption with fluctuating renewable supply • Better utilization of existing network infrastructure, reducing peak load stress, enabling business expansion without requiring expensive infrastructure investment and making better usage of sustainably generated energy • Improved energy security and operational predictability through coordinated consumption patterns • Enhanced trust in digital solutions, reducing time and legal complexity for multi-party data sharing agreements and increasing likelihood of organizations participating in collaborative digital initiatives The impact of the project on the longer term and its societal impact is: • Contributing to Korea's strategic autonomy by improving energy efficiency and reducing vulnerability from energy import dependence (84%) • Supporting Korea's carbon neutrality 2050 commitment through better integration of distributed renewable energy sources and reduced reliance on fossil fuel backup generation • Enabling broader participation in the data economy through deployed data space infrastructure applicable to predictive maintenance, supply chain optimization, and government data sharing beyond energy optimization • Demonstrating that competing organizations can collaborate on critical challenges without sacrificing competitive confidentiality, providing replicable model for other sectors and geographic regions • Advancing confidentiality-preserving technologies and AI-assisted negotiation as enablers for collaborative solutions addressing energy security, climate change, and infrastructure resilience

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Harrie Bastiaansen is a senior researcher at TNO and Endowed Professor at the University of Groningen. His career started with a strong technical background, resulting in a PhD in applied mathematics (1994) when working for KPN Research in the (still promising) area of photonics / integrated optics. Since then, his research focus shifted towards applying major technical system innovations to societal or business benefits. As such, Harrie has had the opportunity to (1) contribute to the development and introduction of KPN's Internet / IP infrastructure, services and processes as converged future-proof basis for all KPN's telecommunication services, both working for KPN Research (as researcher and project lead of various Internet / IP and ATM data networking infrastructure projects) and for KPN Telecom (as manager of the department defining KPN Telecom's strategy and architecture for the fixed data and IP network), (2) the importance of positioning of Enterprise Architecture as part of the corporate strategy, as research topic within TNO and vice-chair of the European ETIS (the Community of European Telecommunication Professionals, www.etis.org) Enterprise Architecture working group, and (3) the role of IT-governance and -auditing as basis for alignment of business and IT-alignment.
Further building upon his core research interest, since 2017 Harrie's main focus has been on the importance of data and data sharing as essential part of the evolving data economy and the role of the EU Data Strategy, with its ambition expressed as the 'Common European Data Spaces', as enabler of the data economy, both within Europe and within the Netherlands. His specific interests are on the legal and governance aspects for the data economy, addressing the main (business and technological) challenges and barriers for large-scale adoption of the emerging federated data sharing concepts.

Freek Bomhof (MSc) is senior consultant at TNO within the Data Science department. He graduated in 1990 from the University of Twente on the topic of Digital Pattern Recognition. He then worked as a researcher, project manager and programme manager at KPN Research on sorting systems that read handwritten addresses on postal letters. From 1997 he was (deputy) R&D manager at the System Integration & Multimedia department. In 2003 he joined TNO as senior project manager, mainly on projects where complex business processes are supported with innovative ICT; notable projects include the introduction of self-steering for engineering teams within KPN including an innovative information system to support this social innovation.

Freek became more interested in the way individuals and organisations adopt ICT innovations, and what role these can play for aspects like sustainability, or how these can help SMEs to stay competitive. He has developed and led numerous projects on the topic of ICT and Sustainability, for instance he was involved as research coordinator in the first phase of the Amsterdam Smart City programme. Freek has worked for many SMEs and SME associations, mainly the engineering sector and building & construction sector. He has led a large European research project on the application of BIM in healthcare design.
Since 2013 Freek works in the Data Science department at TNO, concentrating on applications of data science and Artificial Intelligence in the Defence, Safety & Security domain (and also in other domains), both in early research, collaborative international R&D (EU H2020, ITEA) and close-to-market contract research. With a background in data science, Freek can identify opportunities for applications of research results, while at the same time leveraging his long experience in other fields to make connections to other relevant domains. Multidisciplinarity is the key to the success of any innovation, and this is maybe even more true for data-driven innovations.

Freek is an evaluator for Horizon 2020 project proposals, teaches IPMA internally (D level), is active as taskforce lead in the Big Data Value Association and is a regular and enthusiastic speaker at national and international conferences.

Maarten Kollenstart is a senior scientist within the Data Ecosystems department. He holds a MSc in Computing Science from the University of Groningen. His research focuses on bridging conceptual architectures with concrete implementations, ensuring that data spaces are both secure and practically deployable.
Maarten has played a key role in the technical design of the Smart Connected Supplier Network (SCSN), contributing to its development as a scalable and interoperable data-sharing solution. His expertise spans multiple areas beyond generic data space architecture. From DevOps, with extensive knowledge on Kubernetes and cloud-native architectures, to security with self-sovereign identities, full-stack integrity, and remote attestation.
Maarten led the open-sourcing efforts of the TNO Security Gateway, a comprehensive suite of components for building operational data spaces. The first version of this TSG achieved IDS Assurance Level 2 certification, demonstrating its compliance with the security and trust standards.
Maarten is actively involved in the development of key open standards for Data Spaces. He is a committer to the Eclipse Dataspace Protocol and the Eclipse Decentralized Claims Protocol, both of which are shaping the future of data spaces and are intended for submission to ISO for standardization. He is also a member of the IDSA Working Group Architecture and Gaia-X Working Group Architecture.

Matthijs Punter is a senior consultant in the Data Ecosystems department of TNO. His research focuses on large scale interoperability and digitalization in various industries. He combines state of the art knowledge on new IT concepts with the ability to reach consensus in dynamic business networks.

Internationally he is member of the architecture group of the Industrial Data Space Association, an international cross-industries consortium focusing on controlled data sharing for new data driven business propositions. He is involved in working groups of Gaia-X, the European federated cloud for data sharing.

Within the European Data Spaces Support Centre he has a leadership position in setting-up a blueprint for Common European Data Spaces and its alignment with other initiatives such as SIMPL Open. Within the Netherlands Matthijs is part of the Smart Industry team, an industry initiative to stimulate digitalization in Dutch industry. Examples include supply chain collaboration and the digitalization of manufacturing processes through digital platforms.